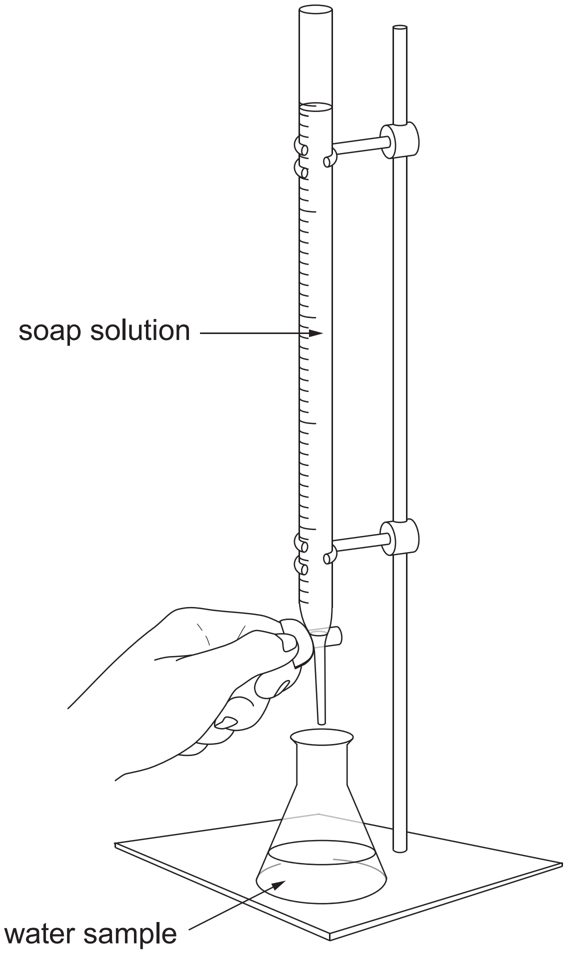
6. Olivia investigated the relative hardness of three samples of water, **A**, **B** and **C**.



She added soap solution to each sample separately and recorded the volume of soap solution needed to get a permanent lather. Her results are shown in the table.

Water sample	Volume of soap solution needed to get a permanent lather (cm^3)
A	1.0
B	14.0
C	11.0

Examiner only

Describe how you would repeat this investigation, including enough detail to show that you will be carrying out a fair test. Assuming that you get similar results to Olivia, state and explain the conclusion you would reach. [6 QER]

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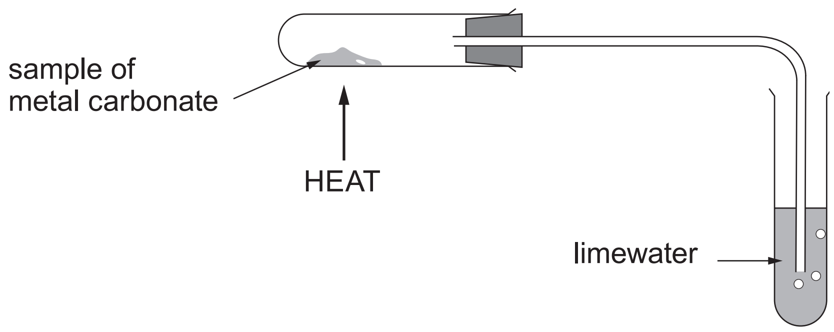
Examiner only

6. *(a)* A student investigated the decomposition of three different metal carbonates.

She measured the time taken for limewater to turn milky using the following apparatus.



Three samples of each metal carbonate were tested. Her results are shown in the table.

Metal carbonate	Time taken for limewater to turn milky (s)			
	Sample 1	Sample 2	Sample 3	Mean
copper(II) carbonate	15	25	17	....
zinc carbonate	54	52	53	53
calcium carbonate	195	200	190	195

(i) Calculate the mean time taken for limewater to turn milky on heating copper(II) carbonate. **Show your working.** [2]

Mean time = ... s

Examiner only

(ii) I. Place the carbonates in order of stability giving a reason for your answer. [2]

Most stable ....

....

Least stable ....

Reason ....

....

II. Explain the order of stability of the carbonates. [1]

....

....

(iii) Complete the following symbol equation for the decomposition of copper(II) carbonate. [2]

$CuCO_3 \longrightarrow$ .... + ...

(b) Calculate the relative formula mass, M_r, of copper(II) carbonate, $CuCO_3$. [2]

$A_r(\text{C}) = 12$ $A_r(\text{O}) = 16$ $A_r(\text{Cu}) = 63.5$

M_r = ...

9

6. This question is about the development of the Periodic Table.

In 1890, not all the elements had been discovered. A Russian scientist named Dmitri Mendeleev used the reactions of the known elements and their relative atomic masses to arrange them in the following table.

I	II	III	IV	V	VI	VII	VIII		
H 1.01									
Li 6.94	Be 9.01	B 10.8	C 12.0	N 14.0	O 16.0	F 19.0			
Na 23.0	Mg 24.3	Al 27.0	Si 28.1	P 31.0	S 32.1	Cl 35.5			
K 39.1	Ca 40.1		Ti 47.9	V 50.9	Cr 52.0	Mn 54.9	Fe 55.9	Co 58.9	Ni 58.7
Cu 63.5	Zn 65.4			As 74.9	Se 79.0	Br 79.9			
Rb 85.5	Sr 87.6	Y 88.9	Zr 91.2	Nb 92.9	Mo 95.9		Ru 101	Rh 103	Pd 106
Ag 108	Cd 112	In 115	Sn 119	Sb 122	Te 128	I 127			
Ce 133	Ba 137	La 139		Ta 181	W 184		Os 194	Ir 192	Pt 195
Au 197	Hg 201	Ti 204	Pb 207	Bi 209					
			Th 232		U 238				

Mendeleev's table

Based on the properties of the known elements, he predicted that new elements would be discovered and left gaps for these in his table. He even predicted what the properties of these undiscovered elements would be. He gave the name ekasilicon to one of them. The element that fits into this gap was eventually discovered and named germanium.

	Predicted properties of ekasilicon (Ek)	Actual properties of germanium (Ge)
Atomic mass	72	72.59
Density (g / cm^3)	5.5	5.35
Melting point	high	937.4 °C
Colour of metal	grey	grey-white
Formula of oxide	EkO_2	GeO_2
Formula of chloride	$EkCl_4$	$GeCl_4$

Comparison of the properties of ekasilicon and germanium

The modern day Periodic Table is based on Mendeleev's original table.

Examiner only

(a) (i) Put a tick (✓) in the correct column to show whether the following statements apply to Mendeleev's table only, today's table only or to both tables. [2]

	Mendeleev only	Today only	Both tables
the table is organised into groups			
copper and potassium are in the same group			
there are gaps in the table			
fluorine and chlorine are in the same group			

(ii) Tick (✓) the **two** statements that **best** describe why germanium was confirmed to be the element ekasilicon predicted by Mendeleev. [1]

germanium has exactly the same atomic mass as that predicted for ekasilicon ☐

germanium has a different colour to that predicted for ekasilicon ☐

germanium has a similar density to that predicted for ekasilicon ☐

germanium oxide has the same ratio of atoms as that predicted for ekasilicon oxide ☐

germanium oxide and germanium chloride have the same ratio of atoms ☐

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Examiner only

(b) (i) Calculate the percentage of oxygen present in the formula of germanium oxide, GeO_2. [2]

$A_r(Ge) = 73$ $A_r(O) = 16$

Percentage = ... %

(ii) When germanium oxide reacts with hydrochloric acid it produces germanium chloride and water.

Balance the following equation for the reaction that takes place. [1]

$$GeO_2 + \boxed{}\,HCl \longrightarrow GeCl_4 + \boxed{}\,H_2O$$

6

Examiner only

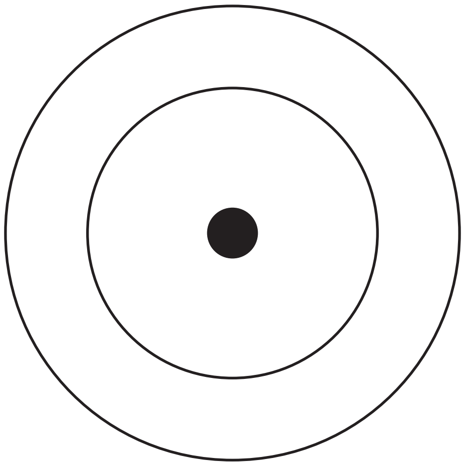
6. (a) An atom of an element has 5 protons, 5 electrons and 6 neutrons.

(i) State the atomic number and the mass number of this atom. [2]

Atomic number ....

Mass number ....

(ii) Complete the diagram of the electronic structure of the element. [1]



(iii) Explain why the atom has no overall charge. [2]

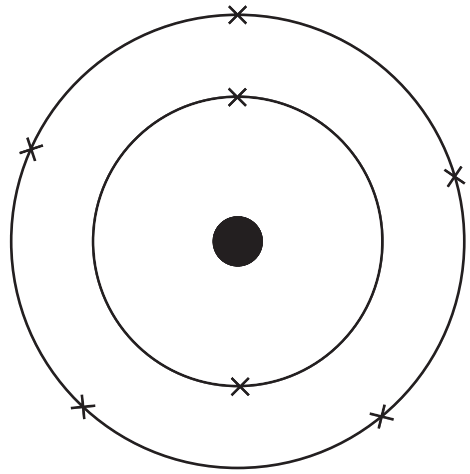
....

....

....

Examiner only

(b) The diagram shows the electronic structure of another element.



This element is in Group 5 and Period 2 of the Periodic Table.

(i) State the name of the element. [1]

....

(ii) Use the electronic structure to explain why this element is in Group 5. [1]

....

....

(iii) Use the electronic structure to explain why this element is in Period 2. [1]

....

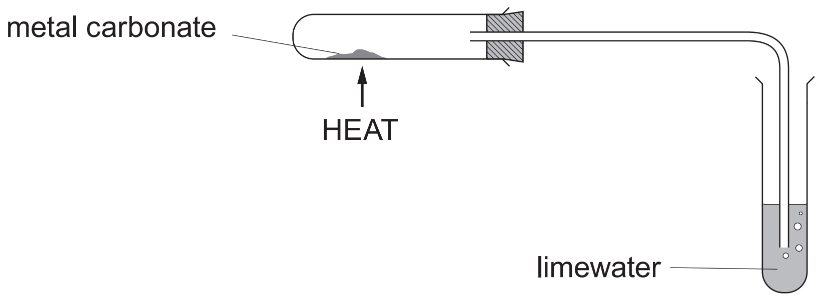
....

8

Examiner only

6. (a) Rhian investigated the decomposition of three different metal carbonates.

She measured the time taken for limewater to turn milky using the following apparatus.



Her results are shown in the table.

Metal carbonate	Time taken for limewater to turn milky (s)
copper(II) carbonate	18
zinc carbonate	27
lead carbonate	11

(i) Place the carbonates in order of stability. [1]

Most stable ....

....

Least stable ....

Examiner only

(ii) If sodium carbonate was used in the investigation the limewater would not turn milky however long it was heated.

Tick (✓) the reason why the limewater would not turn milky. [1]

Sodium carbonate only decomposes a small amount on heating	☐
Sodium carbonate is very unstable	☐
Sodium carbonate does not decompose on heating	☐
Sodium carbonate decomposes too quickly	☐

(iii) On heating copper(II) carbonate, Rhian expected to make 5.0 g of copper(II) oxide. She actually made 3.5 g.

Use the formula below to calculate the percentage yield of copper(II) oxide in her experiment. [2]

$$\text{percentage yield} = \frac{\text{actual mass}}{\text{expected mass}} \times 100$$

Percentage yield = ... %

(iv) One of the ions present in copper(II) carbonate is CO_3^{2-}. [1]

Give the formula of the other ion present.

....

(b) Rhian carried out a flame test to show that sodium carbonate contains sodium ions.

Give the colour of the flame seen. [1]

....

6

